

Scaling Memcached with RCS and CTI

Makayla Coleman and Kate Vaughan

Advisor: Dr. Unnati Shah

1. INTRODUCTION

Memcached challenges

Poor User Experience: Slower page loads and response times lead to a frustrating experience for users, affecting retention and satisfaction.

High Latency: Fetching data from disk-based databases is much slower than retrieving it from memory, resulting in delayed responses.

Scalability Issues: As the number of users grows, the database becomes a bottleneck, making it harder to scale applications efficiently.

Memcache

Memcache is a distributed caching system that helps improve the performance and scalability of applications by storing frequently accessed data in memory.

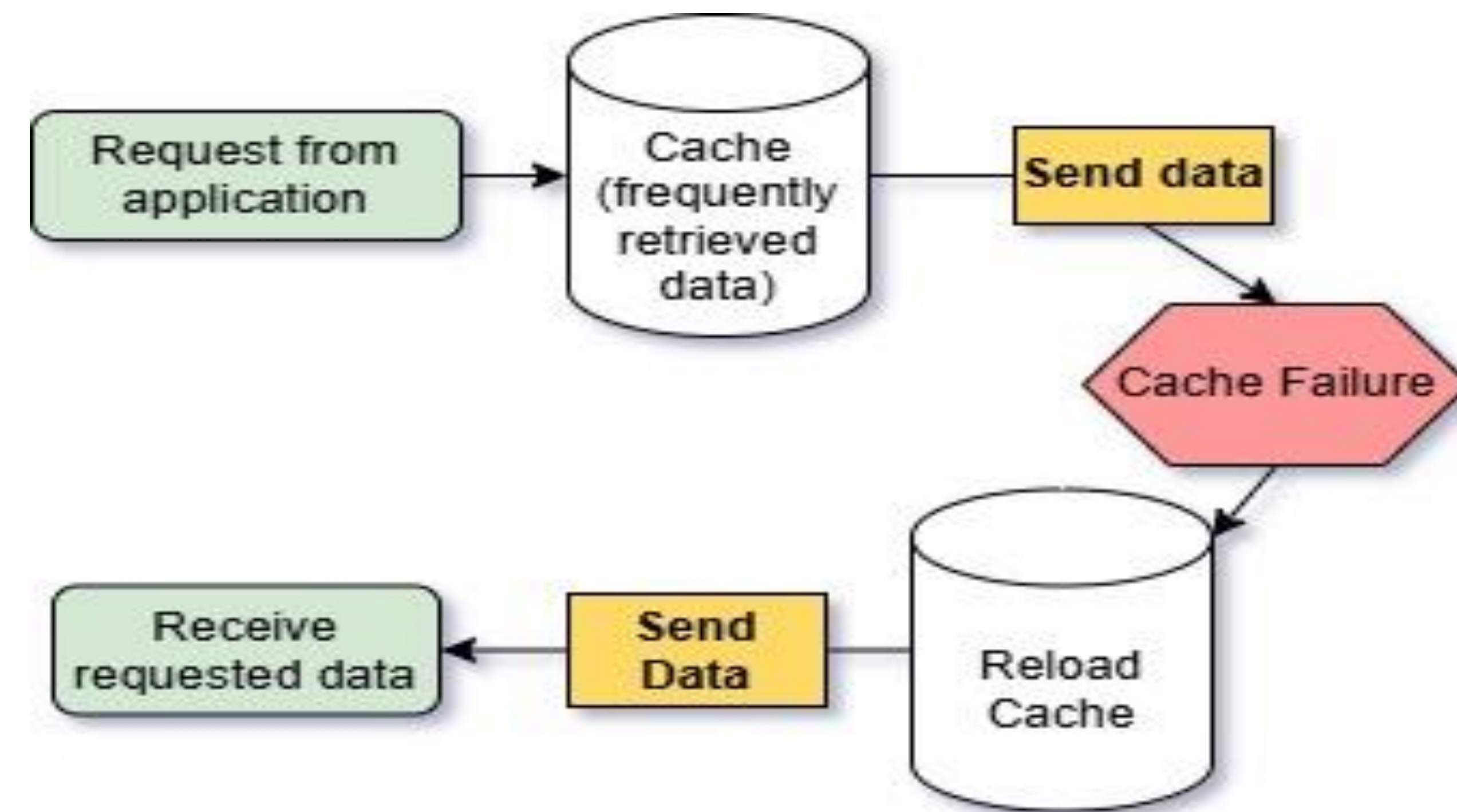


Diagram Represents Regular Cache Failure and Recovery

2. RESEARCH QUESTIONS

1. What is a cache failure and some solutions?
2. What is Resilient Caching Strategies (RCS) and CTI?
3. How does Resilient Caching Strategies (RCS) and Cache Transaction Integrity (CTI) help with Memcached caching failures?
4. What is the difference between regular cache failures and cache failures with Resilient caching Strategies (RCS) and Cache Transaction Integrity (CTI), if any?
5. What RCS did we implement?

3. PROPOSED SOLUTION

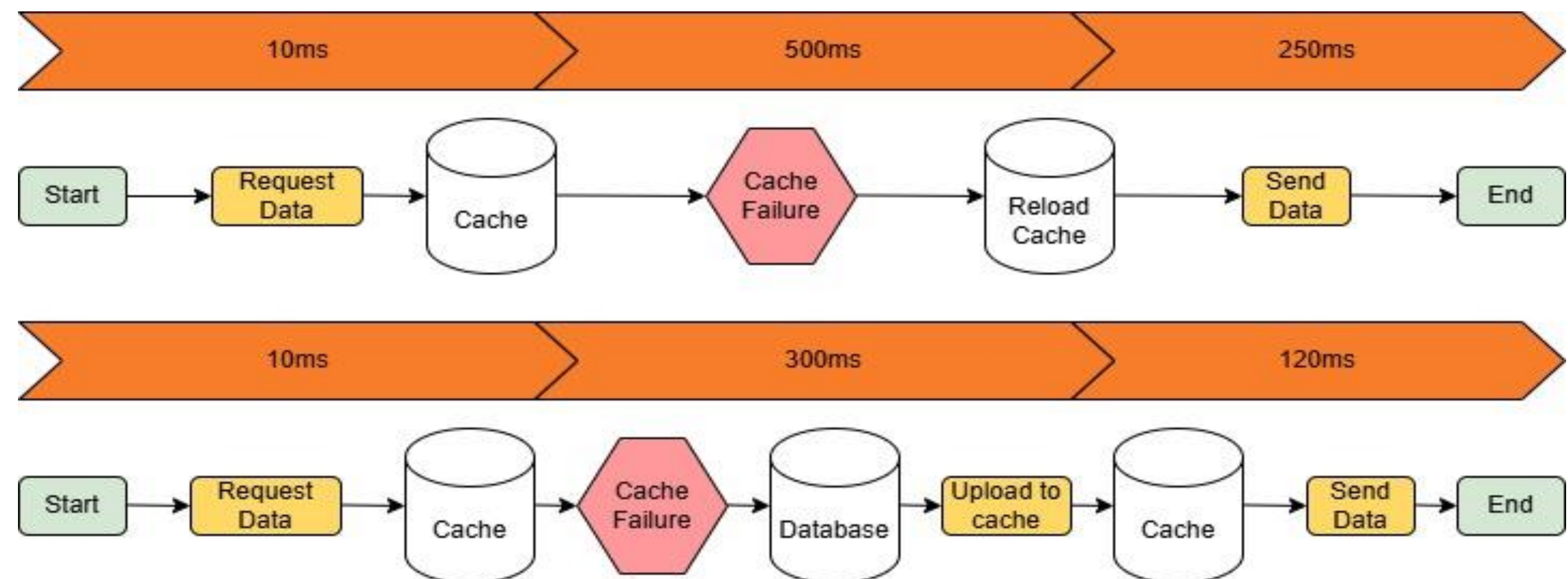
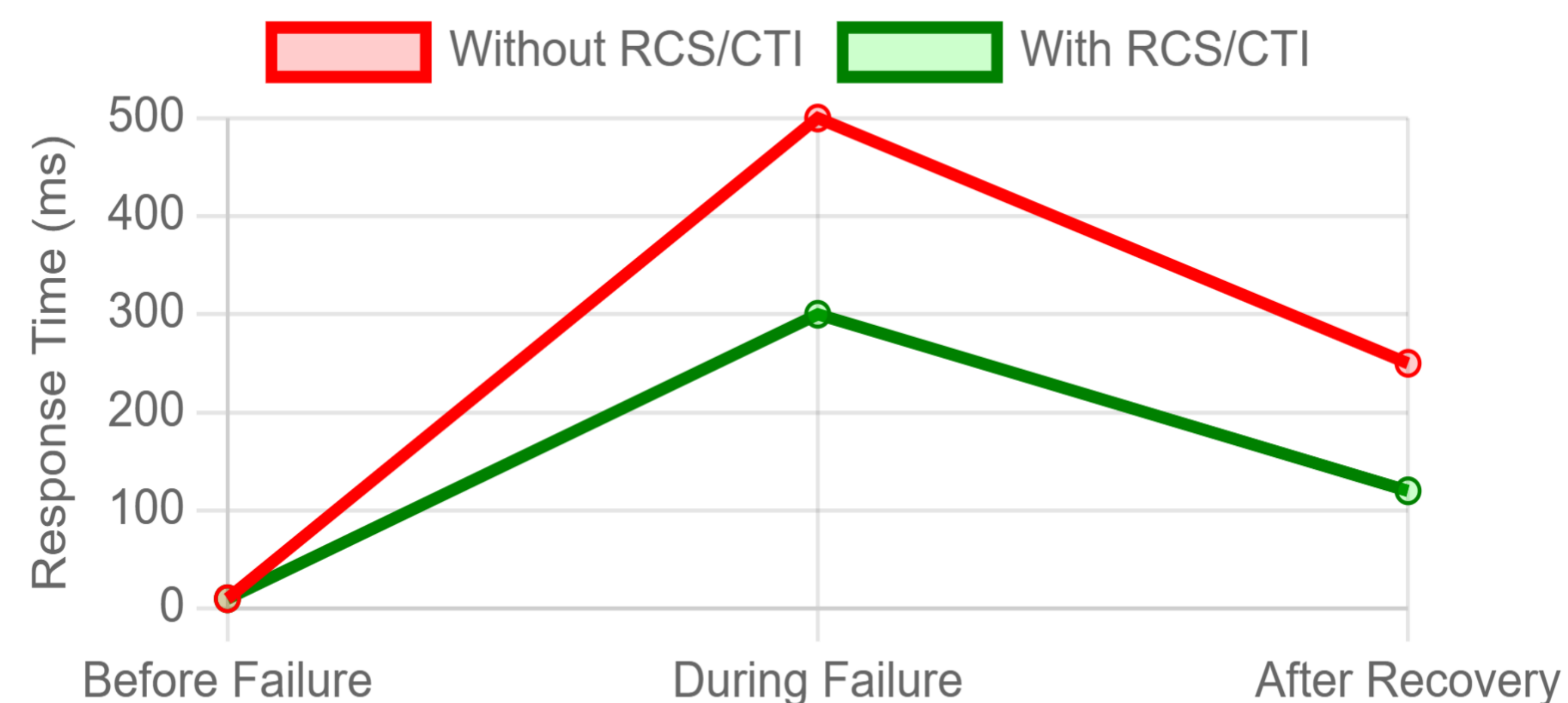


Diagram Represents Regular Memcached System Compared to Memcached with Fallback Approach

4. PRELIMINARY RESULTS



Graph Representing Speed Difference

Scenario	Before Failure	During Failure	After Failure
Without Caching	10ms(Same)	500ms(1.67x slower)	250ms(2.08x slower)
With RCS/CTI	10ms(same)	300ms(1.67x Faster)	120ms(2.08x Faster)

5. CONCLUSION

- Our results demonstrate that integrating Resilient Caching Strategies and Cache Transaction Integrity significantly reduces latency and maintains consistent response times.
- Without RCS/CTI, cache failures result in dramatic performance drops with response times taking much longer than with RCS and CTI incorporated.
- Our implementation using the Fallback strategy shows a 1.6x to 2.1x faster system during and after failures. RCS methods not only enhances stability but also ensures scalable and resilient distributed caching architectures, ultimately mitigating the adverse impacts of sudden load surges and inconsistencies.
- If we continue with this research we hope for the push of using recovery caching methods in the industry to enhance systems' resilience and adaptability to caching failures.

6. REFERENCES

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